Finding Maclaurin's series

Basic skills

Q1. a) $y = 4x^2 - 5$

Poly $x^n \to nx^{n-1}$

$\frac{dy}{dx} = 8x$

"chain" with linear sub

b) $y = \cos 2x$ $\quad \frac{dy}{dx} = -2\sin 2x$

Product

c) $y = e^x \sin x$ $\quad \frac{dy}{dx} = e^x \sin x + e^x \cos x$

d) $y = \tan(x^2)$ $\quad \frac{dy}{dx} = 2x \sec^2(x^2)$

e) $y = \frac{x^2}{\ln x}$ $\quad \frac{dy}{dx} = \frac{\ln x \times 2x - x^2 \times \frac{1}{x}}{\ln^2 x}$

Quotient

$$= \frac{x(2\ln x - 1)}{\ln^2 x}$$

f) $y = \ln^2 x$

$$\frac{dy}{dx} = \frac{1}{x} \times 2\ln x = \frac{2\ln x}{x}$$

Q2. $f(x) = \sin x \quad f'(x) = \cos x$

$f'(0) = \cos 0 = 1$ ← grad at $(0,0)$

Q3. $f(x) = x^2 e^{5x} \quad f'(x) = 2xe^{5x} + 5x^2 e^{5x}$

$2xe^{5x} + 5x^2 e^{5x} = 0$

$\Rightarrow e^{5x}(2x + 5x^2) = 0 \Rightarrow 2x + 5x^2 = 0$

$\neq 0$

$x(2 + 5x) = 0 \Rightarrow x = 0$ or $x = -\frac{2}{5}$

To classify $f''(x) = 2e^{5x} + 10xe^{5x} + 5(2xe^{5x} + 5x^2 e^{5x})$

$f''(0) = 2 + 0 = 2 > 0$ so min

$f''(-\frac{2}{5}) = -0.27 < 0$ so max

Q4. $f(x) = \sqrt{8 + e^x} \quad f'(x) = e^x \sqrt{8 + e^x}$

$f''(x) = e^x \sqrt{8 + e^x} + e^x(e^x \sqrt{8 + e^x})$

$= e^x \sqrt{8 + e^x}(1 + e^x)$

can reuse $\sqrt{8+e^x} \to e^x \sqrt{8+e^x}$ again

Competency skills

Q5. $f(x) = \cos(x^2)$

a) $f'(x) = -2x\sin(x^2)$

$f''(x) = -2\sin(x^2) - 4x^2\cos(x^2)$

b) $f'(0) = 0$, $f''(0) = 0$

c) so need $f(0) = 1$

so $f(x) = 1 + 0x + \frac{0x^2}{2}$

next term is of x^4

Q6. a) $f(x) = e^{3x}$ $\quad f(0) = 1$

$f'(x) = 3e^{3x}$ $\quad f'(0) = 3$

$f''(x) = 9e^{3x}$ $\quad f''(0) = 9$

so $f(x) = 1 + 3x + \frac{9}{2}x^2 + \dots$

b) $f(x) = \tan x$ $\quad f(0) = 0$

$f'(x) = \sec^2 x$ $\quad f'(0) = 1$

+chain

$f''(x) = 2\sec^2 x \tan x$ $\quad f''(0) = 0$

so $f(x) = 0 + 1x + \frac{0x^2}{2} + \ldots$

$= x + \ldots$

c) $f(x) = \sqrt{1+x^2}$ $\quad f'(0) = 1$

$f'(x) = 2x \times \frac{1}{2}(1+x^2)^{-\frac{1}{2}}$ $\quad f''(0) = 0$

$f''(x) = (1+x^2)^{-\frac{1}{2}} + x \times 2x \times -\frac{1}{2}(1+x^2)^{-\frac{3}{2}}$

so $f''(0) = 1$

so $f(x) = 1 + 0x + \frac{1}{2}x^2 + \ldots$

$= 1 + \frac{1}{2}x^2 + \ldots$

Q7. $f(x) = xe^{3x}$

by Q6.

$= x\left(1 + 3x + \frac{9}{2}x^2 + \ldots\right)$

$= x + 3x^2 + \frac{9}{2}x^3 + \ldots$

would achieve same with

$f'(0) = 1, \ f''(0) = 6, \ f'''(0) = 27$

EXAM PAPERS PRACTICE

Advanced skills

Could evaluate these

Q8. $f(x) = \sin(e^x)$ so $f(0) = \sin(1)$

$f'(x) = e^x \cos(e^x)$ so $f'(0) = \cos(1)$

$f''(x) = e^x \cos(e^x) + e^x e^x x - \sin(e^x)$

so $f''(0) = \cos(1) - \sin(1)$

so $f(x) = \sin(1) + \cos(1)x + \frac{(\cos(1) - \sin(1))}{2} x^2 + \ldots$

Q9. $f(x) = \arctan(1+x)$ $f(0) = \frac{\pi}{4}$

$f'(x) = \frac{1}{2 + 2x + x^2}$ $f(0) = \frac{1}{2}$

For derivative $y = \tan^{-1}(1+x)$

$x + 1 = \tan y$

$\frac{d}{dx}(x+1) = \frac{d}{dx}(\tan y)$

$1 = \frac{dy}{dx} \sec^2 y$

so $\frac{dy}{dx} = \frac{1}{\sec^2 y} = \frac{1}{1+\tan^2 y} = \frac{1}{1+(x+1)^2}$

$= \frac{1}{2+2x+x^2}$

so $f(x) \approx \frac{\pi}{4} + \frac{1}{2}x$

Q10. A Maclaurin's series expresses a non-polynomial function as a polynomial, so can be useful for limits and analysis comparing to polynomials (and other functions with maclaurin's series)

Q11. $f(x) = \cos x \qquad f(0) = 1$

$f'(x) = -\sin x \qquad f'(0) = 0$

$f''(x) = -\cos x \qquad f''(0) = -1$

$f'''(x) = \sin x \qquad f'''(0) = 0$

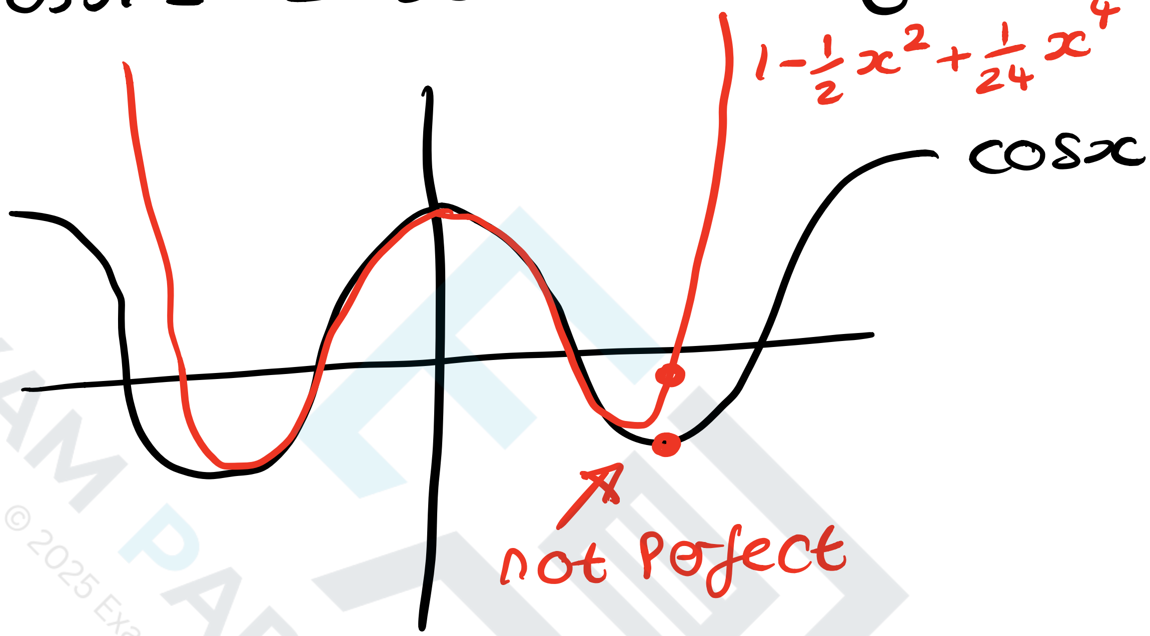
$f''''(x) = \cos x \qquad f''''(0) = 1$

so $f(x) \approx 1 + 0x - \frac{1}{2}x^2 + 0x^3 + \frac{1}{4!}x^4$

$= 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$

so $\cos\pi \approx 1 - \frac{1}{2}\pi^2 + \frac{1}{24}\pi^4 \approx 0.1239...$

well $\cos\pi = -1$ so accuracy is low



More terms helps to improve accuracy away from zero.

Q12. $f(x) = (1-x)^{-1}$ $\quad f(0) = 1$

$f'(x) = -1 \times -1(1-x)^{-2}$ $\quad f'(0) = 1$

$= (1-x)^{-2}$

$f''(x) = 2(1-x)^{-3}$ $\quad f''(0) = 2$

$f'''(x) = 6(1-x)^{-4}$ $\quad f'''(0) = 6$

$f(x) = 1 + 1x + \frac{2}{2}x^2 + \frac{6}{3!}x^3$

$$f(x) = 1 + x + x^2 + x^3$$

$$g(x) = \ln(1+x) \qquad g(0) = \ln 1 = 0$$

$$g'(x) = \frac{1}{1+x} \qquad g'(0) = 1$$

$g'(x) =$ "same" as f

$$g''(0) = 1$$

$$g'''(0) = 2$$

$$g(x) = 0 + 1x + \frac{1}{2}x^2 + \frac{2}{6}x^3$$

$$= x + \frac{1}{2}x^2 + \frac{1}{3}x^3 + \dots$$

so

$$h(x) = f(x)g(x)$$

$$= (1 + x + x^2 + x^3)\left(x + \frac{1}{2}x^2 + \frac{1}{3}x^3\right)$$

$$= x + \frac{1}{2}x^2 + \frac{1}{3}x^3 + x^2 + \frac{1}{2}x^3 + x^3$$

$$= x + \frac{3}{2}x^2 + \frac{11}{6}x^3$$